Normalized Nonlinear Density Comparison on a Gauge Chart

Separating action oscillation from high-dimensional coverage

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Abstract

We prove an exact event comparison between a reference Gaussian measure and a nonlinear perturbation on the same retained chart. The logarithmic loss separates into remainder oscillation and negative log chart mass. A centered Hessian bound H on a radius-r convex ball contributes at most Hr^2. For the side-two four-dimensional block certificate, this local action cost is only 6.35×10^{-6} at the frozen time. By contrast, the directly certified 408-dimensional inner Gaussian ball has log mass -803.25, giving a loss 3.7188 per plaquette after division over 216 plaquettes. Thus local nonlinear geometry is quantitatively adequate, while normalization and chart coverage remain the unresolved bridge.

1 Normalized comparison theorem

Let ν be a probability measure, let D be measurable with $\nu(D) > 0$, and let ν_D be ν conditioned on D. Define

$$\frac{\mathrm{d}\mu_D}{\mathrm{d}\nu_D}(x) = \frac{e^{-R(x)}}{Z_D}, \qquad Z_D = \int_D e^{-R}\,\mathrm{d}\nu_D.$$

Theorem 1.1 (Oscillation and coverage)**.** *If* $\operatorname{osc}_D R \le \kappa$*, then for every event* A*,*

$$\mu_D(A) \le e^{\kappa}\nu_D(A) \le \exp\big(\kappa - \log \nu(D)\big)\nu(A). \tag{1}$$

The analogous lower comparison on events contained in D *is* $\mu_D(A) \ge e^{-\kappa}\nu_D(A)$*.*

Proof. Let $m = \inf_D R$ and $M = \sup_D R$. Then $e^{-M} \le Z_D \le e^{-m}$, so the normalized density ratio lies between $e^{-(M-m)}$ and e^{M-m}. This proves the same-chart comparison. Finally, $\nu_D(A) = \nu(A \cap D)/\nu(D) \le \nu(A)/\nu(D)$. □

Corollary 1.2 (Centered Hessian certificate)**.** *Suppose* D *is contained in a convex radius-*r *ball centered at zero,* $R(0) = 0$*,* $\mathrm{d}R(0) = 0$*, and* $\|\,\mathrm{d}^2 R\| \le H$ *on that ball. Then*

$$|R(x)| \le \frac{H}{2}\|x\|^2, \qquad \operatorname{osc}_D R \le Hr^2.$$

Hence the total log comparison loss in (1) *is at most* $Hr^2 - \log \nu(D)$*.*

Proof. Apply Taylor's formula with integral remainder on the segment from zero to x. The absolute range lies inside $[-Hr^2/2, Hr^2/2]$. □

This formula prevents a common normalization mistake: a pointwise-small remainder does not compare normalized measures unless the reference mass of the retained domain is also controlled.

2 Gaussian chart mass

For $X \sim N(0, tI_n)$,

$$\mathbb{P}(\|X\| \le r) = P\left(\frac{n}{2}, \frac{r^2}{2t}\right),$$

where P is the regularized lower incomplete gamma function. The executable certificate evaluates its logarithm by a positive series in the deep lower tail, avoiding floating-point underflow.

The Stage 08 finite-block input has 136 scalar physical curvature directions and three colors, hence $n = 408$. The certified local radius is $r = 0.003$. Re-evaluating the action certificate at $t = 3 \times 10^{-6}$ gives $H = 0.7051332859$.

quantity	value
dimension n	408
plaquettes	216
inner radius r	0.003
action oscillation Hr^2	6.34620×10^{-6}
inner-ball log mass	-803.2526600
total loss per plaquette	3.71876234
declared target per plaquette	0.2

The inner Euclidean ball is a certified subset of the local coordinate domain, so its mass is a valid but potentially very pessimistic coverage lower bound. Solving the abstract Gaussian problem gives radius 0.02463 to meet the loss target and radius 0.03657 for log mass -0.1. These numbers are diagnostics: the nonlinear chart is not certified at those radii.

3 Frontier consequence

The action remainder is not the dominant obstruction in this certificate. One needs a gauge-covariant atlas or enlarged block chart with adequate total reference mass, controlled overlap multiplicity, transition Jacobians, and no unjustified global gauge slice. Only after those data are supplied can (1) feed a volume-uniform adaptive block estimate.

The theorem is exact on its stated domain but supplies neither full-measure control nor RG, continuum, or spectral conclusions. Its claim level is C2.